



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 31.07.2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : ME F443/ ME U443
Course Title : Quality Control, Assurance and Reliability
Instructor-in-Charge : Prof. Tribeni Roy

1. Course Description:

Basic concepts of probability and probability distribution, Standard probability distribution, Sampling and Sampling Distributions, Confidence intervals, testing significance, Statistical tolerance, Various types of control charts, Statistical process control techniques, Defect diagnosis and prevention, Basic concepts of reliability, Reliability design evaluation and control, Method of applying total quality management into production process.

2. Scope and Objective of the Course:

This course presents an exposition to the modern thoughts on quality and related practices. The principles and techniques for quality control, based on statistical methods and the procedures for quality assurance are covered. Tools for quality engineering and quality management are dealt in detail. Concepts of reliability and methods to improve product and systems reliability are also covered. The course is tailored to enable students to become successful managers in a sustained manner, thereby provide support to the business houses in their never-ending quest for quality.

3. Textbooks:

T1: Mitra A., Fundamentals of Quality Control and Improvement, Wiley Publications, 4th Edition, 2016.

4. Reference Books:

R1: Montgomery, D. C., Introduction to Statistical Quality Control, John Wiley & Sons, 6th Edition, 2009.

R2: Kapur, K. and Lamberson, L., Introduction to Reliability Engineering, John Wiley & Sons, 2nd Edition, 1989.

R3: Montgomery, D.C., Design and Analysis of Experiments, John Wiley & Sons, 8th Edition, 2013.

5. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
M1	L1 – L6: Basic concepts of probability and probability distribution, Standard probability distribution	Class notes	Overview of probability distribution
M2	L7 – L13: Sampling and Sampling Distributions, Confidence intervals, Testing significance, Statistical tolerance.	T1, class notes	Understanding tolerances and sampling distributions



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

M3	L13 – L27: Various types of control charts, Statistical process control techniques, Defect diagnosis and prevention.	T1, class notes	To learn about quality monitoring of processes and diagnosing defects
M4	L28 – L40: Basic concepts of reliability, Reliability design evaluation and control, Method of applying total quality management into production process.	T1, class notes	Understand the basics of reliability and application of total quality management principles

6. Evaluation Scheme*:

Component	Duration	Weightage (%)	Date & Time	Nature of component
Mid-Semester Test	90 mins	30%	8/10 AN1	Open book
Quiz/ Seminar/ Project/ Assignment/ Report etc.	--	35%	To be announced	OB/CB
Comprehensive Examination	180 mins	35%	12/12 FN	Closed book

*This course is compulsory for HD students - *M.E. (Mechanical)*. Therefore 35 % mark is kept for Assignments, Project Presentations and Reports etc.

7. Chamber Consultation Hour: To be announced in the first class.

8. Attendance Policy: Each student is expected to attend all classes and participate in all evaluation components.

9. Notices: All notices related to this course will be communicated via email.

10. Make-up Policy: Make-up will be granted ONLY in genuine cases with prior permission.

11. Minimum Passing Criteria: Student who won't attend mid-semester /comprehensive examination will be awarded NC. Further, the passing marks for this course will be determined based on one of the two criteria: **30% of the median of the post-comprehensive examination score or 20% of the average of the first three top scorers. The lower value between these two criteria will be considered as a passing mark in this course.**

Instructor-in-charge
ME F443/ME U443